Read the quote before calculating

An exchange rate is a price with two currencies attached to it. Before multiplying or dividing, write down what one unit of the base currency costs in the quote currency. This small step prevents most FX calculation errors.

Quote	Meaning
EURUSD = 1.12	EUR 1 costs USD 1.12
GBPUSD = 1.27	GBP 1 costs USD 1.27
USDJPY = 149.50	USD 1 costs JPY 149.50

In EURUSD, EUR is the base currency and USD is the quote currency. If EURUSD rises from 1.12 to 1.15, the euro appreciates against the dollar: buying one euro now takes more dollars. With USDJPY, an increase instead means that the dollar appreciates against the yen. The direction depends on the quote.

$$\text{USD} = \text{EUR} \times \frac{\text{USD}}{\text{EUR}}$$

Suppose you have USD 1,000 and want euros at EURUSD = 1.12. Divide 1,000 by 1.12 to obtain EUR 892.86. Going back to dollars requires multiplication. In a frictionless example, the round trip returns the original USD 1,000.

The dealer buys at the bid and sells at the ask

A quote of EURUSD 1.1198 / 1.1202 means that the dealer buys euros at USD 1.1198 and sells euros at USD 1.1202. If you sell euros, you receive the bid. If you buy euros, you pay the ask. A real round trip therefore loses the spread even when market prices do not change.

> **Keep the viewpoint fixed**
> Bid and ask are stated from the dealer's perspective. Appreciation and depreciation must name both currencies; a currency cannot appreciate in isolation.

Cross rates and a complete arbitrage cycle

A cross rate is the exchange rate implied by two other quotes. Because EURUSD and GBPUSD are both expressed in dollars, dividing them cancels the dollar and gives pounds per euro.

$$\text{EURGBP} = \frac{\text{EURUSD}}{\text{GBPUSD}} = \frac{1.12}{1.27} = 0.881890$$

Suppose the direct market quote is EURGBP = 0.8740. The euro is cheaper in the direct pound market than in the cross rate. Buy euros with pounds, then sell the euros for dollars. To measure profit, begin and end in the same currency.

Step	Conversion	Balance
Start	Initial funds	USD 1,000,000.00
1	USD / 1.27	GBP 787,401.57
2	GBP / 0.8740	EUR 900,917.13
3	EUR x 1.12	USD 1,009,027.19

The round trip earns USD 9,027.19 before transaction costs. Notice the second division: EURGBP is pounds per euro, so pounds divided by pounds per euro produces euros. Reversing that operation changes the transaction.

$$\frac{1}{1.27} \times \frac{1}{0.8740} \times 1.12 > 1$$

Why the gap tends to close

Buying the underpriced euro and selling it through the more expensive route puts pressure on the inconsistent quotes. Two routes to the same final asset should have the same effective cost. This is the no-arbitrage condition.

Use executable prices
This example uses one price per pair to isolate the logic. With bid-ask quotes, use the correct side on every trade, account for fees and funding, and verify that the whole round trip still produces a profit.

What moves FX, and what arbitrage does not explain

Arbitrage explains consistency across simultaneous prices. It does not tell us which exchange rate will prevail tomorrow. Exchange rates change when investors revise their expectations about returns, inflation, growth, risk and future currency demand.

Interest rates matter through relative returns

A higher domestic interest rate can make domestic assets more attractive, but the effect is not automatic. If the higher rate reflects inflation or financial stress, investors may still avoid the currency. What matters is the expected return after currency movements and risk, relative to the alternatives.

Trade and investment create currency demand

An importer needs foreign currency to pay a supplier. A pension fund may need it to purchase securities abroad. These transactions connect exchange rates to the real economy and portfolio decisions. A company can be exposed on both sides: foreign sales create receipts, while foreign purchases create payments.

Carry trades retain exchange-rate risk

A carry trade borrows in a lower-yielding currency and invests in a higher-yielding currency. The interest differential is not a guaranteed profit. An adverse exchange-rate move can erase the interest income, especially when the position is leveraged or investors unwind it quickly. Covering that currency risk brings the forward exchange rate into the calculation, which is the subject of Lec05.

Price consistency is different from market integrity

A market can have closely aligned cross rates and still suffer from misconduct. The 2015 U.S. foreign-exchange enforcement cases concerned coordination among traders in the USD/EUR market. Arbitrage across currency pairs does not by itself prevent collusion or abusive benchmark trading.

> **The habit to carry forward**
> Identify the quote units, the currency you own, the currency you need, and the cash-flow date. Follow the money through the complete transaction before deciding whether there is a gain.

Case reference: U.S. Department of Justice, May 20, 2015, "Five Major Banks Agree to Parent-Level Guilty Pleas."

justice.gov/archives/opa/pr/five-major-banks-agree-parent-level-guilty-pleas